Auteur: Yoren Collier
Studierichting: Informatica
Oefeningenreeks 4A nr. 38.2

$$I = \int \frac{2x+7}{\left(x^2+1\right)^2} dx = \int \frac{2x}{\left(x^2+1\right)^2} dx + 7 \int \frac{1}{\left(x^2+1\right)^2} dx$$

Stel: $t = x^2 + 1 \Rightarrow dt = 2xdx$

$$= \int \frac{1}{t^2} dt + 7 \int \frac{1}{\left(x^2+1\right)^2 dx}$$

$$= \frac{1}{-1t} + c + 7 \left[\frac{2 \cdot 2 - 3}{2 \cdot 2 - 2} \cdot \int \frac{1}{\left(x^2+1\right)} dx + \frac{x}{\left(2 \cdot 2 - 2\right)\left(x^2+1\right)} \right]$$

$$= \frac{-1}{x^2+1} + c + 7 \left[\frac{1}{2} \cdot \operatorname{Bgtan}(x) + c + \frac{x}{2\left(x^2+1\right)} \right]$$

$$= \frac{-1}{x^2+1} + \frac{7}{2} \operatorname{Bgtan}(x) + \frac{7x}{2\left(x^2+1\right)} + c$$

$$= \frac{-2+7x}{2\left(x^2+1\right)} + \frac{7}{2} \operatorname{Bgtan}(x) + c$$

References